22 $\left(16^{-\frac{3}{4}}\right)^{-2} \times 2^m = 1$

Find the value of m.
You must show all your working.

Fourth root of 16 = 2

$16^{\frac{-3}{4}} = 2^{-3} = \frac{1}{8}$

1 mark
A negative fractional power means reciprocal, root, then power.

$(1/8)^{-2} = 8^2 = 64 = 2^6$

1 mark
The outside power −2 inverts, then squares.

$2^6 \times 2$ to the power $m = 2$ to the power $(6 + m) = 1$

Since $1 = 2^0$, $6 + m = 0$, so $m = -6$.

$m =$ ..... −6 .....

(Total for Question 22 is 3 marks)

1 mark
Equal bases imply equal exponents.

DO NOT WRITE IN THIS AREA
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